



FG-MoE: Heterogeneous mixture of experts model for fine-grained visual classification

Songming Yang^{a,b}, Jing Wen^{a,*}, Bin Fang^a

^a College of Computer Science, Chongqing University, Chongqing, 401331, China

^b School of Computer Science and Engineering, Southeast University, Nanjing, 211189, China

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ABSTRACT

Fine-grained visual classification (FGVC) is a challenging task due to subtle inter-class differences and significant intra-class variations. Most existing approaches struggle to simultaneously capture multi-level discriminative features and effectively integrate complementary visual information. To address these challenges, we propose Fine-Grained Mixture of Expert (FG-MoE), a novel heterogeneous mixture-of-experts model for fine-grained visual classification. Our approach introduces a specialized multi-scale pyramid module that aggregates multi-scale information and enhances feature representation through spatial and channel attention mechanisms. Inspired by neuroscientific insights into visual processing mechanisms of the human brain, FG-MoE employs five specialized experts that focus on different visual cues: global structures, regional semantics, local details, textures, and part-level interactions. A spatial-aware gating mechanism dynamically selects appropriate expert combinations for each input image. We further design a novel multi-stage training strategy and employ balance constraints along with diversity and orthogonality regularization to ensure balanced learning and promote diverse expert specialization. The final classification leverages fused features from all selected experts. Extensive experiments on three widely used FGVC datasets demonstrate that FG-MoE achieves substantial performance improvements over backbone models and establishes state-of-the-art results across all these benchmarks, validating the effectiveness and robustness of our approach.

1. Introduction

Fine-grained visual classification (FGVC) enables precise classification of subcategories within broader categories, such as similar species or brands, with applications in biodiversity monitoring and intelligent transportation [1]. Compared to general image classification, FGVC poses greater challenges due to subtle inter-class differences and significant intra-class variation caused by pose, viewpoint, and background factors.

In recent years, FGVC methods have evolved from CNN-based to Transformer-based. Early CNN-based methods relied on manually annotated part-level information or region proposal networks, but struggled to simultaneously capture global structures and fine-grained local details. Subsequently, Transformer-based methods effectively modeled long-range dependencies through self-attention mechanisms, yet lacked sensitivity to local features [2].

These limitations become particularly pronounced given that FGVC inherently requires analyzing multi-granularity cues such as shape, texture, and part relationships—an area where current single-path and

homogeneous architectures often fall short [3]. Effectively processing these distinct visual cues remains a key bottleneck.

Nowadays, Mixture-of-Experts (MoE) models, which employ a gating network to coordinate specialists, have succeeded in natural language processing (NLP) [4] and general vision tasks [5]. However, the application of MoE architectures to address the specific multi-granularity challenges of FGVC remains relatively underexplored.

Since FGVC distinctions often depend on subtle variations (e.g., shape, component, texture), they demand specialized feature extractors. This naturally motivates our design of a heterogeneous MoE architecture, where specialized experts can be tailored to capture distinct granularity-specific cues critical for fine-grained discrimination.

Thus, We propose FG-MoE, a fine-grained visual classification model combining multi-expert collaboration and competition with adaptive routing and gating. Our framework integrates a multi-scale pyramid (MSP) module with a specific MoE system featuring five specialized experts: global structure, regional attributes, local details, texture patterns, and part interactions. A spatial gating network enables dynamic expert

* Corresponding author.

E-mail addresses: yangsongming321@163.com (S. Yang), wj@cqu.edu.cn (J. Wen), fb@cqu.edu.cn (B. Fang).

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routing, while balancing and diversity regularization ensure specialization without collapse. The main contributions are:

- We propose a specific multi-scale pyramid structure that fuses multi-scale information and applies specialized attention enhancement, enabling the model to focus on discriminative regions and provide rich visual cues for expert processing.
- We construct a heterogeneous MoE system with five specialized experts capturing global semantics, regional contrasts, local details, multi-directional textures, and part-based relationships for complementary feature extraction based on neuroscience theory.
- We design a spatially-aware gating mechanism that dynamically selects optimal expert combinations for each image based on its content, enabling effective processing of FGVC tasks.
- We introduce expert balancing and diversification strategies to prevent degeneration and ensure specialization, addressing the “winner-takes-all” issue while promoting diverse learning.

The rest of this paper is organized as follows. Section 2 reviews related work. Section 3 details the proposed FG-MoE framework. Section 4 presents experimental results and analysis. Section 5 summarizes our work.

2. Related work

In this section, we first review the development of fine-grained visual classification methods, then discuss MoE and the existing applications of MoE in FGVC.

2.1. Fine-grained visual classification

Early research on FGVC was primarily based on CNN. Initial methods depended on manually annotated part-level information [2,3], which resulted in high annotation costs. To reduce this burden, weakly supervised approaches using only image-level labels were proposed. These included Region Proposal Networks for localizing discriminative regions [6], as well as feature encoding methods that learned more expressive representations [7].

Following Transformers’ success in NLP, Vision Transformer [8] extended to computer vision by processing image patches through self-attention mechanisms. Transformer-based methods leverage attention weights to filter key patch tokens and enhance discriminative regions [9,10]. Swin Transformer [11] introduced shifted window attention for local-global feature integration, inspiring improvements like MPFA method [12].

In recent years, researchers have explored various approaches to enhance fine-grained recognition capabilities. The CMD method [13] models the relationship between image patches and fine-grained semantics through a cross-hierarchy multi-instance distillation framework, enhancing self-supervised representation learning. The CGL method [14] decomposes the visual concepts of fine-grained categories into inherited and discriminative concepts, improving recognition robustness in complex scenarios.

While existing approaches often utilize a unified architectural framework, MoE paradigm introduces an alternative strategy. By coordinating specialized subnetworks through dynamic gating mechanisms, MoE models offer enhanced adaptability for complex tasks.

2.2. Mixture of experts model

The Mixture of Experts model, originally proposed by Jacobs et al. [15], is an ensemble learning framework adopting a divide-and-conquer strategy. The MoE architecture consists of expert networks with shared structural design but different parameters, a gating network that dynamically assigns input-dependent weights, and an aggregation mechanism combining expert outputs to enable adaptive computation balancing model capacity and efficiency. Following the success of the

MoE in NLP, researchers have begun extending MoE to computer vision. A notable milestone is V-MoE by Riquelme et al. [5], which applied large-scale sparse MoE to image classification.

2.3. Existing attempts of MoE in FGVC

In fine-grained classification, MoE applications remain exploratory. Zhang et al. [16] proposed a multi-granularity expert framework, where each expert specializes at a particular scale. The ME-ASN method [17] constructs multi-granularity training data through attention exchange and progressive augmentation strategies, introducing attention mechanisms to enhance the mixture of experts model and improve expert diversity and fine-grained recognition capability.

However, current MoE-based methods face several challenges. First, most adopt static expert assignment, lacking dynamic adaptability to complex scenes. For instance, MGE’s [16] expert collaboration is limited, making it hard to capture intra-class variation. Second, many focus on specific feature types, ignoring the integration of global and local cues. Third, inconsistent expert outputs lack coordination, hindering overall performance.

Structurally, homogeneous experts with simplistic gating and poor layer-wise fusion limit the modeling of diverse, multi-level fine-grained features. To address these issues, we propose a heterogeneous MoE that leverages enhanced multi-scale features from our MSP module. Our approach tackles FGVC challenges through dynamic gating and routing, supported by balance and diversity regularization.

3. Method

In this section, we present an introduction to the proposed Fine-Grained Mixture of Experts (FG-MoE) model. The model employs a multi-scale pyramid module to extract and fuse hierarchical feature information, which is subsequently fed into heterogeneous expert modules and a gating network. The gating network intelligently assigns classification tasks to different expert modules for processing, while incorporating diversity and balance constraints. Through this dynamic expert allocation strategy, the model ultimately achieves effective fine-grained image classification. Meanwhile, a multi-stage training strategy ensures effective and stable operation of the model.

3.1. Framework

This paper proposes a fine-grained visual classification model based on heterogeneous Mixture of Experts. The overall architecture of the model is shown in Fig. 1. The model consists of three core components: feature extraction backbone, multi-scale feature pyramid module, and MoE system.

When an image is fed into the model, the pre-trained backbone feature extraction network (pretrained Swin Transformer [11], Vision Transformer [6], ConvNeXt [18], ResNet [19] etc. For descriptive convenience, we use Swin Transformer to represent the backbone throughout this paper) first extracts multi-stage features from the input image. We select features from specific stages as multi-scale inputs and feed them into the multi-scale pyramid (MSP) module. The MSP module performs dimensionality reduction and fusion of the multi-scale features from the backbone, and further enhances spatial and channel attention through the enhanced spatial-channel attention (ESCA) module. The enriched multi-scale features, after fusion and enhancement by MSP, are then fed into both the gating network and the five experts of the MoE system for further processing. The gating network assigns the optimal expert combination for each image, employing hard routing during training and soft routing during testing. Meanwhile, balance constraints and diversity regularization jointly ensure the stability and richness of the expert system. Finally, the fused results from various experts’ feature extraction help us determine the final classification. The entire model

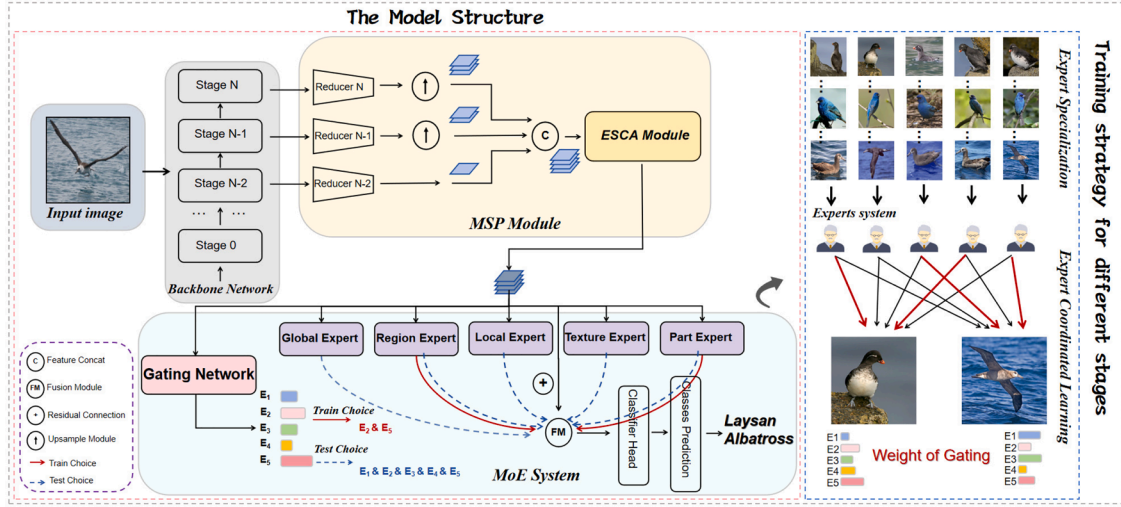


Fig. 1. Framework of FG-MoE model.

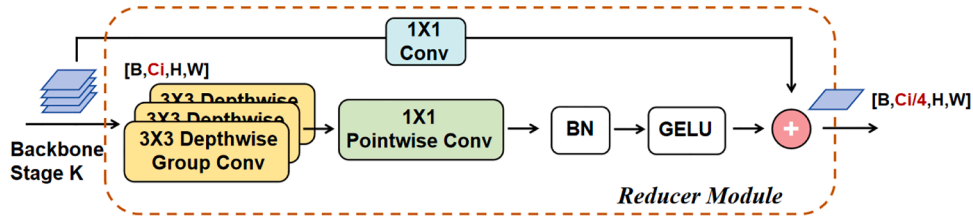


Fig. 2. Reducer module.

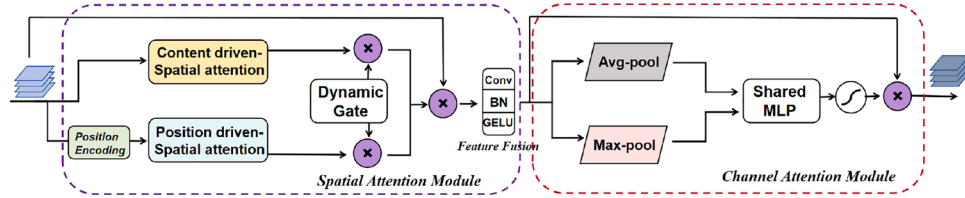


Fig. 3. ESCA module.

is trained using a multi-stage strategy. The subsequent subsections will elaborate on these aspects in detail.

3.2. Construction of multi-scale feature pyramid

The MSP module constructs rich multi-scale representations from backbone features, bridging different network stages and integrating features with varying semantic levels and spatial resolutions. Through feature compression, upsampling, and ESCA module, the MSP provides unified feature representations for the MoE system.

Given input image $I \in \mathbb{R}^{H \times W \times 3}$, we extract hierarchical feature maps $F_i = \text{Backbone}_i(I)$ from backbone networks such as Swin Transformer. Each feature map F_i captures representations at different abstraction levels and spatial resolutions.

3.2.1. Reducer module

As shown in Fig. 2, the Reducer Module employs depthwise separable convolution for efficient channel dimensionality reduction of backbone multi-scale features. It sequentially performs spatial feature extraction via 3×3 depthwise group convolution (groups = C_i), 1×1 pointwise

convolution for cross-channel fusion and channel compression to $C_i/4$, followed by batch normalization and GELU activation. With residual connections, the module reduces computational overhead while preserving spatial resolution, providing an effective foundation for multi-scale feature fusion.

Following dimensionality reduction via the Reducer module, the processed features are spatially aligned through upsampling operations and concatenated along the channel dimension, before being fed into the ESCA module for additional feature enhancement.

3.2.2. ESCA module

As shown in Fig. 3, ESCA module integrates spatial and channel attention mechanisms for refined multi-scale feature enhancement. The spatial attention employs both content-driven and position-driven mechanisms, with content-driven attention generating spatial weights from feature content and position-driven attention learning spatial importance through position encoding. A dynamic gate adaptively fuses these two spatial attentions using learnable parameters. The channel attention module adopts a dual-pathway design with global average pooling and max pooling to extract channel-level information, processed

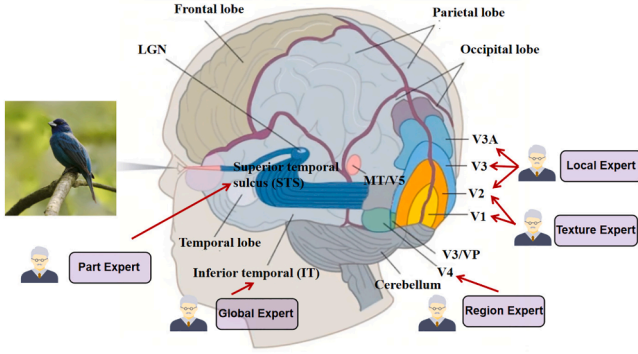


Fig. 4. Corresponding relationship between experts and the human brain visual system.

through a shared MLP. The design of channel attention here partially follows the idea of CBAM [20]. Finally, the module outputs enhanced feature representations with precise spatial localization and channel selectivity for subsequent expert networks.

This MSP module thus provides a rich, unified feature representation by integrating semantic abstraction and spatial detail from multiple scales. It forms the foundational input for the subsequent MoE system, enhancing its ability to reason across visual granularity levels.

3.3. The design of heterogeneous experts in FG-MoE.

The MoE system, which constitutes the core innovation of this approach, comprises five specialized experts for feature extraction, integrated with a dynamic routing mechanism. Each expert is dedicated to capturing different specific type of feature pertinent to fine-grained classification, collectively forming a comprehensive feature extraction framework. This section elaborates on the design principles and implementations of each expert module.

3.3.1. Neuroscience-inspired expert design

Our multi-expert mixture model is inspired by the hierarchical processing of the human visual system. Visual information flows through the ventral pathway for object classification and dorsal pathway for spatial processing [21], forming an ideal computational reference for fine-grained visual classification.

The five-expert design computationally references and extends this biological structure through specific mappings: the global expert corresponds to the IT region for object classification and category integration; the region expert maps to V4/PIT for moderately complex features and inter-regional relationships; the local expert corresponds to V2/V3 for edge combinations and local shapes; the texture expert maps to V1/V2, mimicking simple and complex cell responses to texture and orientation; and the part relationship expert corresponds to STS and the dorsal pathway for spatial configurations and structural relationships. The approximate correspondence between relevant brain regions and the design of the expert modules is illustrated in Fig. 4.

Similar to dynamic visual processing in humans, our MoE gating network simulates this regulation by dynamically allocating expert weights based on input features, to allow autonomous pathway selection for different samples.

3.3.2. Global expert

The Global Expert captures holistic structural information and global semantic features in fine-grained images. While local details are crucial, the overall morphology, pose, and structural layout provide essential discriminative clues.

As shown in Fig. 5(A), this expert models long-range dependencies between all spatial positions in the image through an 8-head multi-head

attention mechanism [22]. This global modeling capability enables the model to perceive complete object structures, preventing over-focus on local details while neglecting holistic features. The feature enhancement module further refines global semantic representations through non-linear transformations (expanding dimensions by $2\times$ then compressing), while the channel attention mechanism (We reference the method of SENet [23]) adaptively selects the most valuable global feature dimensions for the current classification task. Ultimately, the enhanced global features are fused with original features, providing macro-level discriminative information for fine-grained classification.

3.3.3. Region expert

The Region Expert focuses on decomposing fine-grained images into semantically meaningful regions and modeling inter-region and comparative relationships. Notably, this expert effectively considers foreground-background relationships, extracting richer contextual information.

Specifically, as shown in Fig. 5(B), the region extractor first extracts local features through convolutional operations, then the spatial attention module generates N (N is set to 4) soft attention masks to partition the feature map into N semantic regions. These masks are learned end-to-end through a lightweight convolutional network that outputs N probability maps, where each map highlights a distinct semantic region through spatial softmax normalization. This soft partitioning strategy allows the model to autonomously learn the most discriminative region segmentation rather than relying on predefined part detection. After processing each region independently, we compute contrastive features between all region pairs ($\binom{N}{2}$ pairs), capturing subtle differences through contrastive learning—particularly crucial for distinguishing similar species. The contrastive features are fused and further enhanced through a region attention mechanism and spatial attention gating to strengthen discriminative information, ultimately producing region-aware feature.

3.3.4. Local expert

The Local Expert specializes in capturing local detail features. This expert extracts discriminative local features through multi-scale convolutions and edge detection.

As shown in Fig. 5(C), the expert employs three parallel local feature extraction branches, using grouped convolutions with dilation rates of 1, 2, and 3 respectively. This multi-scale design enables the model to capture fine details (small receptive field) and local structures (large receptive field). The extracted multi-scale features are fused and enhanced through a detail enhancement module, which refines the most discriminative local features via progressive channel compression and expansion.

Simultaneously, the edge detector uses pre-initialized Sobel operators to detect object contours and boundaries, which are crucial for identifying shape characteristics of fine-grained categories. The local attention mechanism further highlights important local regions. Finally, the enhanced detail features are fused with edge features, generating locally-sensitive feature representations.

3.3.5. Texture expert

The Texture Expert extracts and analyzes texture patterns and directional features in fine-grained images. Texture information serves as crucial cues for distinguishing similar species—such as feather texture direction, density, and regularity in birds.

As shown in Fig. 5(D), two sets of Gabor filters [24] with 3×3 and 5×5 kernels correspond to different spatial frequencies. The Gabor filter is defined as: $g(x, y; \lambda, \theta, \sigma, \gamma) = \exp\left(-\frac{x'^2 + y'^2}{2\sigma^2}\right) \cos\left(\frac{2\pi x'}{\lambda}\right)$ where $x' = x \cos \theta + y \sin \theta$, $y' = -x \sin \theta + y \cos \theta$, θ is the orientation (8 directions), λ is wavelength, σ is Gaussian envelope standard deviation, and $\gamma = 0.5$ is spatial aspect ratio.

The high-frequency enhancement module dynamically adjusts texture response intensity through a learnable contrast parameter. Meanwhile, Sobel edge detector and Laplacian operator capture first-order

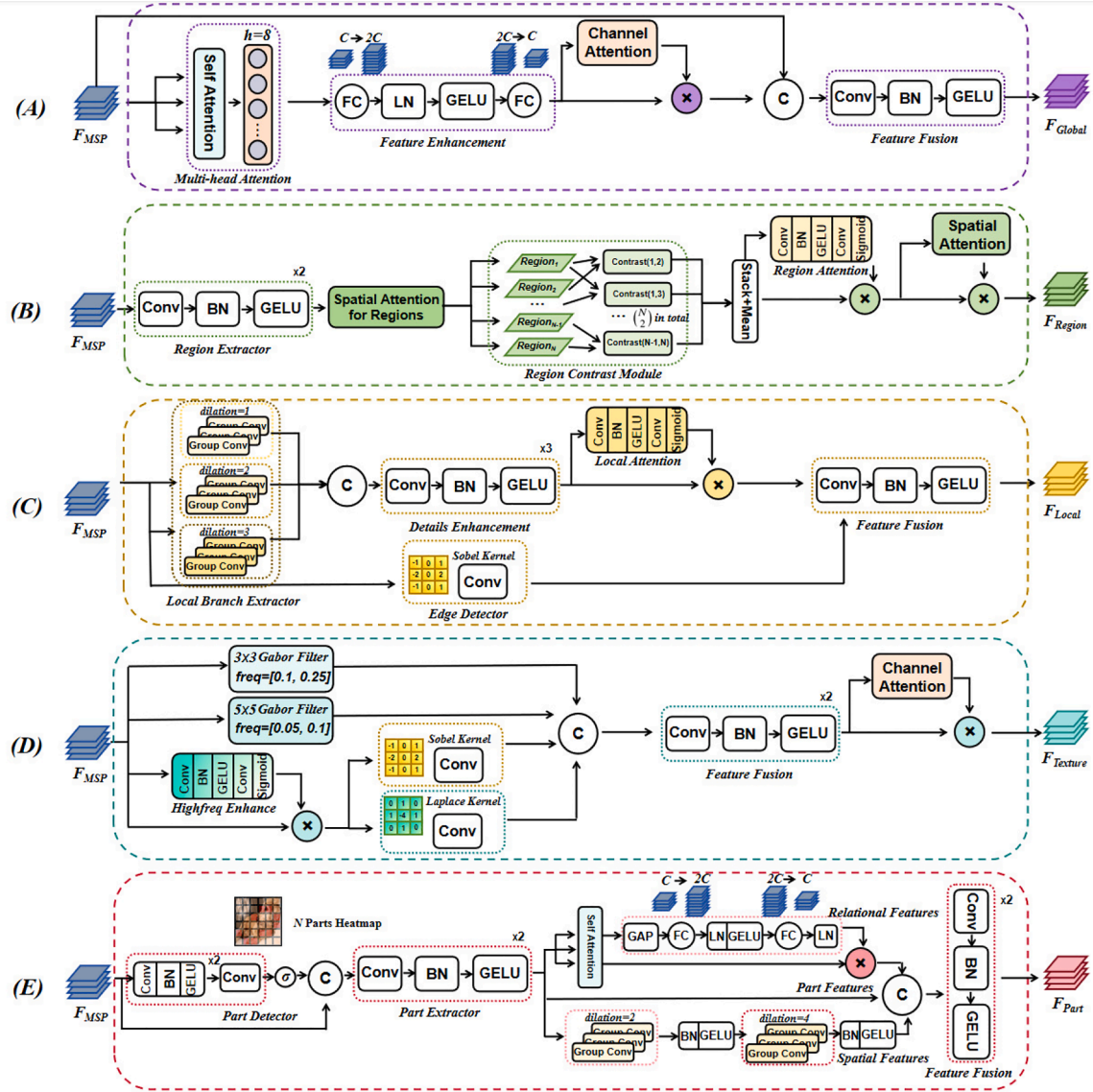


Fig. 5. The detailed architecture of 5 experts.

and second-order texture boundary information respectively. All texture features are concatenated and fused, then modulated through an adaptive channel attention mechanism, ultimately generating texture-sensitive feature representations.

3.3.6. Part expert

The Part Expert is dedicated to identifying and modeling spatial relationships and semantic connections between key parts in fine-grained images.

As shown in Fig. 5(E), this expert first generates heatmaps for $N(N$ is set to 8) key parts through a part detector, where each heatmap represents the spatial probability distribution of the corresponding part. The part detector consists of convolutional layers, with each output channel learning to activate on a semantically meaningful part through end-to-end training, without requiring explicit part annotations. These heatmaps are concatenated with original features and processed through a part feature extractor. The core innovation lies in the triple relation modeling mechanism: (1) The Non-Local Block captures long-range dependencies between different parts through self-attention [22], with query and key dimensions compressed to $C/2$ for computational efficiency;

(2) The spatial relation module uses grouped convolutions with dilation rates of 2 and 4 to capture inter-part spatial relationships at different scales; (3) The relation reasoning module infers abstract relationships between parts in high-level semantic space through global average pooling and a two-layer fully connected network.

The three types of relational features (part features, relational features, and spatial features) are ultimately fused through an aggregation network, generating feature representations that encode rich part relationship information.

3.4. Design of the spatially-aware gating strategy

The gating mechanism serves as the control core of the MoE system, dynamically determining expert activation and assigning contribution weights based on input content. This adaptive routing strategy enables flexible adjustment of feature extraction approaches according to different image characteristics, effectively addressing sample diversity challenges in fine-grained visual classification.

Traditional MoE systems typically use simple gating networks based on global pooling features, which ignore spatial information crucial for

expert routing in fine-grained classification. For instance, prominent bird head features should activate the part expert more, while clear feather textures should rely more on the texture expert.

We propose a spatial context-aware gating mechanism that extracts both global and local information to provide comprehensive expert selection. The spatial context encoder processes input feature maps from the MSP module through global average pooling for global features and spatial grid features through adaptive pooling to 4×4 grids, enabling location-sensitive expert selection. These spatial features are processed by convolutional layers and flattened to spatial representations, which are then fused with global features to construct comprehensive spatial context representation and directly generate expert importance scores:

$$S = \text{GateNetwork}(\text{Fusion}(\text{Concat}[F_{\text{global}}, F_{\text{spatial}}])) \in \mathbb{R}^{B \times 5} \quad (1)$$

where F_{global} represents global semantic features, F_{spatial} represents 4×4 spatial grid features, and S denotes the importance scores for five experts. The mechanism employs dynamic adaptation with different routing strategies during training and testing while maintaining lightweight efficiency despite enhanced spatial capabilities.

Ultimately, expert outputs are fused through adaptive weights generated by the gating network:

$$F_{\text{MoE}} = \sum_{i=1}^5 w_i \cdot F_{E_i} \quad (2)$$

where w_i represents the dynamic weight of the i th expert and F_{E_i} denotes the output features of the i th expert. The fused features F_{MoE} are processed through global average pooling, normalization, and fully connected layers to directly output the final classification prediction, ensuring that fine-grained classification decisions rely entirely on the fusion results of multi-expert features. Experiments demonstrate that this spatially-aware approach significantly improves expert selection accuracy compared to traditional global-feature-based gating mechanisms.

3.5. Dynamic expert routing strategy

Once the expert importance scores are obtained, the key challenge in the gating mechanism is determining how to select and activate the appropriate experts. This study proposes a dynamic selection strategy that adjusts the expert routing process based on the differing requirements of the training and testing phases.

During training phase, experts are driven to specialize and diversify, enabling stable gradient propagation and reduced redundancy; during testing phase, they retain richer information for smoother predictions, collectively enhancing model performance.

Hard routing strategy (Training phase) During the training phase, this study implemented a hard routing strategy based on Gumbel-Softmax, which strictly selects k experts for each sample. The gating network first generates logits $S \in \mathbb{R}^{B \times N_e}$ for expert selection, where B is the batch size and N_e is the number of experts. This approach encourages each expert to learn specialized feature representations:

$$\hat{S} = \text{GumbelSoftmax}(S/\tau, \text{hard} = \text{True}) \quad (3)$$

$$W_i = \text{TopK}(\hat{S}, k) \quad (4)$$

where $\hat{S}$ denotes the Gumbel-Softmax output, τ is the temperature parameter that gradually decreases throughout the training process, and W_i represents the binarized expert routing matrix indicating which experts are selected for each sample.

Soft routing strategy (Testing phase) During the testing phase, this study employs a soft routing strategy based on Softmax, allowing multiple experts to contribute simultaneously with different weights:

$$W_i = \text{Softmax}(S/\tau) \quad (5)$$

where S represents the same expert selection logits from the gating network, W_i now represents the continuous expert weight matrix, and τ

serves as a dynamically adjustable temperature parameter used to control the “softness” of expert routing.

The impact of the selection parameter k in hard routing, as well as the strategies adopted during training and testing, will be experimentally analyzed in Section 4.4.

In order to ensure that all experts are adequately and evenly utilized, and to prevent issues such as “expert collapse” (where some experts are hardly used) or “expert homogeneity,” this study specifically designs an expert balance and diversity mechanism.

Balance loss Based on expert usage statistics, an enhanced expert balance loss is designed to promote the balanced use of experts:

$$\mathcal{L}_{\text{balance}} = \lambda_1 \cdot \mathcal{L}_{\text{entropy}} + \lambda_2 \cdot \mathcal{L}_{\text{low}} \quad (6)$$

Here, $\mathcal{L}_{\text{entropy}}$ is the entropy deficit loss based on expert usage frequency, encouraging a uniform distribution by minimizing the gap between maximum entropy and current entropy, while $\mathcal{L}_{\text{low}}$ is a penalty term for experts with low usage, aimed at boosting the usage of experts below a threshold:

$$\mathcal{L}_{\text{low}} = \sum_{i=1}^{N_e} \max(0, \theta - f_i)^2 \quad (7)$$

where θ is the usage rate threshold and f_i represents the usage frequency of expert i .

Diversity loss To ensure that different experts learn complementary semantic feature representations, a diversity loss is introduced to minimize the similarity between expert outputs:

$$\mathcal{L}_{\text{diversity}} = \frac{1}{N_p} \sum_{i < j} \cos(F_i, F_j) \quad (8)$$

where F_i and F_j are the output features of expert i and expert j , N_p is the number of expert pairs, and $\cos$ denotes the cosine similarity.

Orthogonality loss To further ensure the global feature representations of different experts are functionally orthogonal, an orthogonality loss is designed to encourage expert features to be orthogonal in space:

$$\mathcal{L}_{\text{ortho}} = \frac{1}{N_p} \sum_{i < j} \|G_i^T G_j\|_F^2 \quad (9)$$

where G_i and G_j are the expert features after global pooling, and $\|\cdot\|_F$ represents the Frobenius norm.

The total loss function $\mathcal{L}_{\text{total}}$ optimized by the model is expressed as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda_b \cdot \mathcal{L}_{\text{balance}} + \lambda_d \cdot \mathcal{L}_{\text{diversity}} + \lambda_o \cdot \mathcal{L}_{\text{ortho}} + \lambda_{\text{reg}} \cdot \mathcal{L}_{\text{reg}} \quad (10)$$

where $\mathcal{L}_{\text{cls}}$ denotes the cross-entropy loss for classification, $\mathcal{L}_{\text{balance}}$, $\mathcal{L}_{\text{diversity}}$, and $\mathcal{L}_{\text{ortho}}$ are the unique losses of the MoE system. The coefficients λ_b , λ_d , and λ_o control the weights of these auxiliary losses. Additionally, we use L2 regularization $\lambda_{\text{reg}} \cdot \mathcal{L}_{\text{reg}}$ to prevent overfitting, with λ_{reg} set to 0.0002. The optimal values for λ_b , λ_d , and λ_o are explored in subsequent experiments.

Through the collaborative and competitive operation of the five expert modules, the mixed expert system can understand image content from multiple dimensions and abstraction levels, capturing comprehensive representations that range from global semantics to local details, and from texture features to part relationships. This enables powerful and diverse feature extraction for fine-grained visual classification.

3.6. Multi-stage training strategy

Given the complexity of the overall MoE system, effective training strategies are essential. This research therefore establishes two fundamental stages, shown in Fig. 1:

Expert specialization stage The gating network is frozen. Uniform routing distributes inputs evenly among experts by sample index, ensuring equal data exposure. This prevents expert dominance and allows

Table 1
Dataset statistics of CUB-200-2011, Stanford-Dogs and NABirds.

Dataset	Classes	Train	Test
CUB200-2011 [1]	200	5994	5794
Stanford Dogs [25]	120	12,000	8,580
NABirds [26]	555	23,929	24,633

each expert to fully develop its specialization, focusing on distinct fine-grained features.

Expert coordinated learning stage The gating network is unfrozen with a doubled learning rate to accelerate its learning. Dynamic expert selection replaces uniform routing. Gumbel-Softmax hard routing promotes specialization, complemented by adaptive temperature regulation to balance expert utilization.

This multi-stage strategy effectively addresses common MoE issues such as expert homogenization, imbalanced utilization, and unstable gating, enabling the model to fully leverage the MoE system and achieve optimal classification accuracy.

4. Experiments

In this section, to evaluate the performance of our proposed FG-MoE method, we conducted experiments on three widely used benchmark datasets. We first introduce the benchmark datasets, evaluation metrics, and implementation details, then present comparative analysis against state-of-the-art methods, followed by ablation studies and visualization analysis.

4.1. Dataset and metrics

In this validation process, We conducted experiments on three standard fine-grained classification datasets in Table 1: (1) CUB-200-2011 [1] contains 11,788 images of 200 bird species; (2) Stanford Dogs [25] includes 20,580 images of 120 dog breeds; (3) NABirds [26] comprises 48,562 images of 555 North American bird species.

We primarily use Top-1 classification accuracy as the evaluation metric, supplemented by Top-5 accuracy, Precision, Recall, and F1-Score for comprehensive assessment. The number of categories and summary of dataset split are shown in dataset.

4.2. Implementation details

Following standard practice [27,28], our backbone network is initialized with pre-trained weights from ImageNet across all datasets. Training images were resized to 448×448 then randomly cropped to 384×384 , while validation images underwent center cropping. All images were normalized using ImageNet standards. The AdamW optimizer was used with weight decay 0.05 and gradient accumulation steps 8. Batch sizes were 4 for CUB-200-2011 and NABirds, and 8 for Stanford Dogs. All experiments were conducted using PyTorch on a Nvidia RTX4090D-24G GPU.

4.3. Comparison with state-of-the-art methods

In this section, we compare our FG-MoE model with state-of-the-art methods, with the best results in bold and second-best underlined. These results validate the superior capability of our proposed method. Notably, our method uses 384×384 input images, achieving competitive performance with greater computational efficiency than most existing approaches using 448×448 inputs.

Evaluation and analysis on the CUB-200-2011 dataset. As shown in Table 2, our method achieves 93.0% classification accuracy, demonstrating superior performance over existing methods.

Table 2
Comparison results on the CUB-200-2011 dataset.

Method	Publication	Backbone	Resolution	Acc (%)
Cross-X [29]	CVPR 2019	ResNet-50	448×448	87.7
FDL [30]	AAAI 2020	DenseNet-161	448×448	89.1
API-Net [31]	AAAI 2020	DenseNet-161	448×448	90.0
PMG [32]	ECCV 2020	ResNet-50	448×448	88.9
GarD [33]	CVPR 2021	ResNet-50	448×448	89.6
SFFF [34]	TCSVT 2022	ResNet-50	n/a	85.4
PMG-V2 [35]	TPAMI 2022	ResNet-101	448×448	90.0
GDSMP-Net [36]	PR 2023	ResNet-50	224×224	89.9
LGTF [37]	ICCV 2023	DenseNet-161	448×448	91.5
CAMF [27]	SPL 2021	Swin-B	448×448	91.2
RAMS-Trans [10]	AAAI 2021	ViT-B 16	448×448	91.3
ViT-Net [38]	ICML 2022	Swin-B	448×448	91.6
TransFG [9]	AAAI 2022	ViT-B 16	448×448	91.7
DCAL [39]	CVPR2022	ViT-B 16	448×448	91.4
Mix-ViT [40]	PR 2023	ViT-B 16	448×448	91.0
AA-trans [41]	PR 2023	ViT-B 16	448×448	91.4
IELT [42]	TMM 2023	ViT-B 16	448×448	91.8
MP-FGV [43]	AAAI 2024	ViT-B 16	448×448	91.8
MPSA [12]	TIP 2024	Swin-L	384×384	<u>92.9</u>
FN-Net [44]	PR 2025	ViT-B 16	448×448	91.4
TransIFC [45]	TMM 2025	Swin-B	448×448	90.6
TransIFC + [45]	TMM 2025	Swin-B	448×448	91.0
MGIFR-Net [28]	PR 2025	Swin-B	384×384	92.2
Baseline	–	Swin-B	384×384	91.1
		Swin-L	384×384	92.0
FG-MoE (Ours)	–	Swin-B	384×384	92.3
		Swin-L	384×384	93.0

Among Transformer-based methods, our approach achieves performance improvements over TransFG, MP-FGV, and IELT. While these methods rely on single feature extraction strategies or limited attention mechanisms, our MoE system leverages specialized experts to simultaneously capture diverse feature representations. Compared with CNN-based approaches such as Cross-X and PMG, our method achieves significant improvements. Unlike traditional CNN methods with fixed feature extraction processes, our MoE system dynamically selects optimal expert combinations based on input characteristics, enabling more flexible feature extraction. Within Swin architecture-based methods, our approach demonstrates substantial advantages over TransIFC, CAM, and ViT-Net, and outperforms MPSA which also uses Swin-L as backbone. These methods employ single-strategy approaches such as invariant feature extraction or complementary attention fusion, while our MoE system adapts to different fine-grained features through multi-expert collaboration, achieving comprehensive feature representation while avoiding the limitations of fixed feature extraction strategies used in existing approaches.

Evaluation and analysis on the Stanford Dogs dataset. As shown in Table 3, our method achieves 93.7% classification accuracy.

Compared to Transformer-based methods such as RAMS-Trans, TransFG, and IELT, our approach achieves notable improvements. While these methods rely on singular feature extraction strategies, our MoE system leverages specialized experts to capture diverse visual features through dynamic expert allocation. Against CNN-based methods including Cross-X and API-Net, our method demonstrates significant improvements, overcoming CNN limitations in simultaneously capturing global context and local details. Notably, our FG-MoE achieves substantial enhancement over the Swin-L baseline, demonstrating effective performance improvement without increased complexity—particularly valuable for datasets with subtle inter-class differences and partial occlusions.

Evaluation and analysis on the NABirds dataset. As shown in Table 4, our method achieves 92.2% classification accuracy, outperforming existing methods and demonstrating superior performance on

Table 3
Comparison results on the Stanford Dogs dataset.

Method	Publication	Backbone	Resolution	Acc (%)
DSTL [46]	CVPR 2018	Inception-v3	448 × 448	85.2
Cross-X [29]	CVPR 2019	ResNet-50	448 × 448	88.9
MGE-CNN [16]	ICCV 2019	ResNet-50	448 × 448	88.5
FDL [30]	AAAI 2020	DenseNet-161	448 × 448	84.9
API-Net [31]	AAAI 2020	ResNet-101	448 × 448	90.3
PMG-V2 [35]	TPAMI 2022	ResNet-101	448 × 448	90.7
SFFF [34]	TCSVT 2022	ResNet-50	n/a	88.0
RAMS-Trans [10]	AAAI 2021	ViT-B 16	448 × 448	92.4
SIM-Trans [47]	ACM MM 2022	ViT-B 16	448 × 448	91.1
TransFG [9]	AAAI 2022	ViT-B 16	448 × 448	92.3
ViT-Net [38]	ICML 2022	Swin-B	448 × 448	90.3
PIM [48]	arxiv 2022	Swin-B	448 × 448	92.1
AFTTrans [49]	ICASSP 2022	ViT-B 16	448 × 448	91.6
AA-trans [41]	PR 2023	ViT-B 16	448 × 448	90.8
IELT [42]	TMM 2023	ViT-B 16	448 × 448	91.8
MP-FGVC [43]	AAAI 2024	ViT-B 16	448 × 448	91.0
FAL-ViT [50]	TCSVT 2025	ViT-B 16	448 × 448	91.1
Baseline	–	Swin-B	384 × 384	91.0
		Swin-L	384 × 384	92.7
FG-MoE (Ours)	–	Swin-B	384 × 384	92.4
		Swin-L	384 × 384	93.7

Table 4
Comparison results on the NABirds dataset.

Method	Publication	Backbone	Resolution	Acc (%)
DSTL [46]	CVPR 2018	Inception-v3	448 × 448	87.9
Cross-X [29]	ICCV 2019	ResNet-50	448 × 448	86.2
MGE-CNN [16]	ICCV 2019	ResNet-50	448 × 448	88.0
API-Net [31]	AAAI 2020	DenseNet161	448 × 448	88.1
GarD [33]	CVPR 2021	ResNet-50	448 × 448	88.0
PMG-V2 [35]	TPAMI 2022	ResNet-101	448 × 448	88.4
GDSMP-Net [36]	PR 2023	ResNet-50	224 × 224	89.0
LGTF [37]	ICCV 2023	DenseNet-161	448 × 448	90.4
TransFG [9]	AAAI 2022	ViT-B 16	448 × 448	90.8
IELT [42]	TMM 2023	ViT-B-16	448 × 448	90.8
MP-FGVC [43]	AAAI 2024	ViT-B 16	448 × 448	91.0
LDH-ViT [51]	PR 2025	ViT-B 16	448 × 448	90.2
FAL-ViT [50]	TCSVT 2025	ViT-B 16	448 × 448	90.3
TransIFC [45]	TMM 2025	Swin-B	448 × 448	90.8
TransIFC + [45]	TMM 2025	Swin-B	448 × 448	90.9
Baseline	–	Swin-B	384 × 384	90.5
		Swin-L	384 × 384	<u>91.8</u>
FG-MoE (Ours)	–	Swin-B	384 × 384	91.2
		Swin-L	384 × 384	92.2

this large-scale, challenging dataset with numerous visually similar bird species.

Our method shows consistent advantages across different Transformer-based architectures including FAL-ViT, MP-FGVC, TransIFC, TransIFC + and IELT. While these approaches rely on singular strategies such as noise reduction, text guidance, or discriminative region focus, our multi-expert system captures comprehensive visual features essential for bird classification. Against CNN-based methods such as LGTF and GarD, we achieve substantial improvements. Unlike fixed feature extraction strategies, our dynamic expert selection adapts to diverse bird characteristics, enabling more effective feature representation for species with subtle morphological differences.

Comparison on other MoE-based method. To further evaluate the model performance, we compared FG-MoE with other MoE-based models on CUB dataset. Currently, there are relatively few fine-grained visual classification models based on the MoE architecture, with the mainstream models being MGE (ICCV) [16] and ME-ASN (TMM) [17]. Unfortunately, the ME-ASN method did not provide its code, making it im-

Table 5
Comparison on MoE-based method.

Method	Resnet 50(%)	ViT-B 16(%)	Swin-L(%)	Param(base on Swin-L)
MGE(ICCV)	88.5	90.6	92.3	352.94 M
FG-MoE(Ours)	87.8	91.6	93.0	214.54 M

Table 6
Modules ablation experiments on different datasets.

Method	CUB-200-2011(%)	Stanford Dogs(%)	NABirds(%)
Baseline	91.99	92.67	91.79
Baseline + MSP	92.51	93.42	92.05
Baseline + MoE	92.68	93.38	92.09
Baseline + MSP + MoE	92.96	93.68	92.19

Table 7
Experts ablation study on the CUB-200-2011 dataset.

Global	Region	Local	Texture	Part	Acc (%)
✓	✓	✓	✓	✓	92.96
✗	✓	✓	✓	✓	92.62
✓	✗	✓	✓	✓	92.68
✓	✓	✗	✓	✓	92.77
✓	✓	✓	✗	✓	92.72
✓	✓	✓	✓	✗	92.72

possible to conduct a comparison of model performance and parameter quantities across different backbones. Therefore, we chose to compare with MGE by re-implementing its code.

As shown in Table 5, FG-MoE establishes a significant performance advantage over MGE on the Transformer architecture, despite using significantly fewer parameters. However, MGE exhibits superior capability on the CNN architecture. We hypothesize that this is because MGE's expert-gating mechanism is likely designed to more effectively leverage the inherent locality and multi-level hierarchical features of CNN architectures, whereas FG-MoE's attention-based routing is optimized for the global, sequential feature representation of Transformers.

Finally, Fig. 6 presents a comparative analysis between FG-MoE and other Transformer-based models that report results on at least two of the three datasets.

4.4. Ablation study

During the ablation studies, this paper employed the same training configurations as previously described, including learning rate, batch size, and optimizer settings. For each experimental group, a complete training cycle was conducted, and the best accuracy results obtained on the test set were recorded.

Table 6 shows the results of module ablation research. We conducted ablation studies on each module of the model. After removing the MoE system or MSP module, there was a decrease in performance. These results indicate that each component of FG-MoE has made an effective contribution to the overall performance.

Table 7 presents the changes in model accuracy after removing different experts. In this experiment, the training and evaluation processes remained consistent, with the selective disabling of certain expert modules used to observe their impact on overall model performance. The results demonstrate that optimal performance is achieved when all five experts collaborate, with performance degradation observed upon removal of any individual expert. These findings validate the necessity of each expert and their synergistic effects. While individual contributions vary, all experts provide positive enhancements, confirming the effectiveness of our MoE system design.

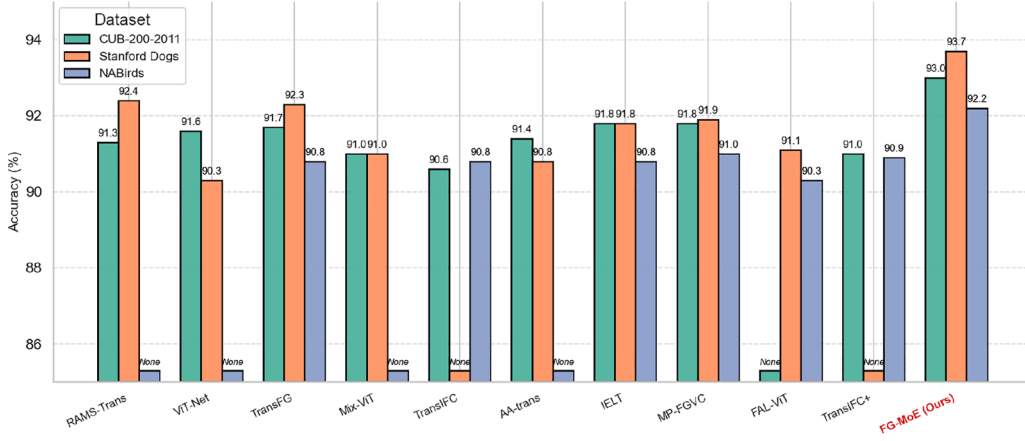


Fig. 6. Comparison on transformer-based methods across multiple datasets.

Table 8

The value of k replacement experiments on the CUB-200-2011 dataset.

Metrics	k = 1	k = 2	k = 3	k = 4	k = 5
Acc(%)	92.53	92.96	92.79	92.82	92.81

Table 9

Comparison of accuracy of different training or testing strategies.

Method	Train Strategy	Test Strategy	Acc (%)
1	Hard	Hard	92.69
2	Hard	Soft	92.96
3	Soft	Hard	92.42 ^a
4	Soft	Soft	92.61 ^a

^a represents instability during training at times

Given that our training employs a Top-K routing mechanism, the value of k critically influences training outcomes. To investigate this effect, Table 8 presents performance evaluations across 5 different k values, along with per-batch average using times during the coordination learning phase (see Section 4.2 for experimental details). The results indicate that although training time scales with increasing k values, $k = 2$ yields the optimal trade-off between performance and computational cost. Therefore, we ultimately chose 2 as the assignment for k .

As shown in Table 9, hard routing demonstrates clear advantages for practical MoE training. While soft routing offers theoretical flexibility by allowing continuous weighting across experts, it introduces significant numerical instabilities—particularly from reciprocal operations—that often cause training collapse via NaN propagation. Conversely, the discrete assignment of hard routing ensures robust training stability, making it the preferred choice in practice.

Since our total loss function $\mathcal{L}_{total}$ includes multiple weight parameters ($\lambda_{balance}$, $\lambda_{diversity}$, and λ_{ortho}) that significantly affect model performance, we conducted loss weight comparison experiments using the Swin-L backbone on the CUB-200-2011 dataset, informed by our empirical practice. Results are shown in Table 10.

Here, the weights of the balancing mechanism ($\lambda_{balance}$) and the diversity mechanisms ($\lambda_{diversity}$ and λ_{ortho}) are configured to exhibit a trade-off relationship. Our experiments demonstrate that the model achieves optimal performance when $\lambda_{balance} = 0.835$, $\lambda_{diversity} = 0.185$, and $\lambda_{ortho} = 0.185$, based on averaged results from multiple trials. It can be observed that model performance does not exhibit significant fluctuations

when these weights are varied, indicating the robustness of our model to parameter changes. Furthermore, substantial performance degradation occurs when either the balancing mechanism or the diversity mechanisms are removed.

To evaluate the universality of our FG-MoE model, we conducted backbone replacement experiments across diverse architectures including ResNet50, ConvNeXt Base, ViT-B/16, Swin-B, and Swin-L on the CUB-200-2011 dataset, as shown in Table 11.

Our method achieves consistent improvements across all tested architectures, validating the excellent compatibility of our multi-expert fusion mechanism with different feature extraction paradigms. Notably, all evaluated models exhibit substantial improvements when integrated with FG-MoE, indicating our method's particular effectiveness for Transformer-based architectures. When combined with Swin-L, our FG-MoE achieves optimal performance of 93.0%, confirming the practical value with state-of-the-art networks.

4.5. Evaluation with multiple metrics

To provide comprehensive performance evaluation beyond standard Top-1 accuracy, we additionally employ Top-5 accuracy, Precision, Recall, and F1-score across all datasets. As shown in Table 12, our method achieves excellent performance across all metrics and datasets.

Notably, Top-5 accuracy exceeds 98.5% on all datasets, demonstrating robust performance even with highly similar categories. The balanced precision-recall metrics indicate effective misclassification reduction while maintaining sufficient correct sample capture, highlighting our method's classification effectiveness and robustness.

4.6. Visualization

In this section, we present various visualizations, including saliency map visualization, expert decision visualization, and confusion matrix visualization.

4.6.1. Multi-perspective visual comparison

As shown in Fig. 7, a comprehensive visual comparison between the baseline Swin-Large and the FG-MoE is conducted from three perspectives: Saliency Map, Activation Map, and Grad-CAM. Swin Transformer-Large exhibits scattered and fragmented attention across multiple image regions. In contrast, our FG-MoE, through multi-expert collaboration and dynamic feature selection, precisely concentrates attention on critical discriminative regions, forming coherent semantic understanding that directly enhances fine-grained classification performance.

Table 10
Comparison on the impact of different loss function weight on model performance.

Parameter	Set 1	Set 2	Set 3	Set 4	Set 5	Set 6	Set 7	Set 8	Set 9	Set 10
$\lambda_{balance}$	1.200	0.835	0.534	0.400	0.320	0.835	0.835	0.835	w/o	w/o
$\lambda_{diversity}$	0.120	0.185	0.264	0.400	0.480	0.185	w/o	w/o	0.185	w/o
λ_{ortho}	0.120	0.185	0.264	0.400	0.480	w/o	0.185	w/o	0.185	w/o
Acc (%)	92.88	92.96	92.85	92.83	92.82	92.69	92.76	92.64	92.69	92.53

Table 11
Backbone replacement experiments on the CUB-200-2011 dataset.

Backbone	Backbone Acc(%)	FG-MoE Acc(%)	Improved(%)
Resnet 50	84.0	87.8	+3.8
ConvNeXt Base	90.5	91.4	+0.9
ViT Base 16	90.3	91.6	+1.3
Swin Base	91.1	92.3	+1.2
Swin Large	92.0	93.0	+1.0

Table 12
Evaluation with multiple metrics on different datasets.

Dataset	Top-1 Acc(%)	Top-5 Acc(%)	Precision	Recall	F1-Score
CUB-200-2011	92.96%	98.55%	0.9328	0.9300	0.9294
Stanford Dogs	93.68%	99.47%	0.9355	0.9350	0.9341
NABirds	92.19%	99.48%	0.9151	0.9057	0.9061

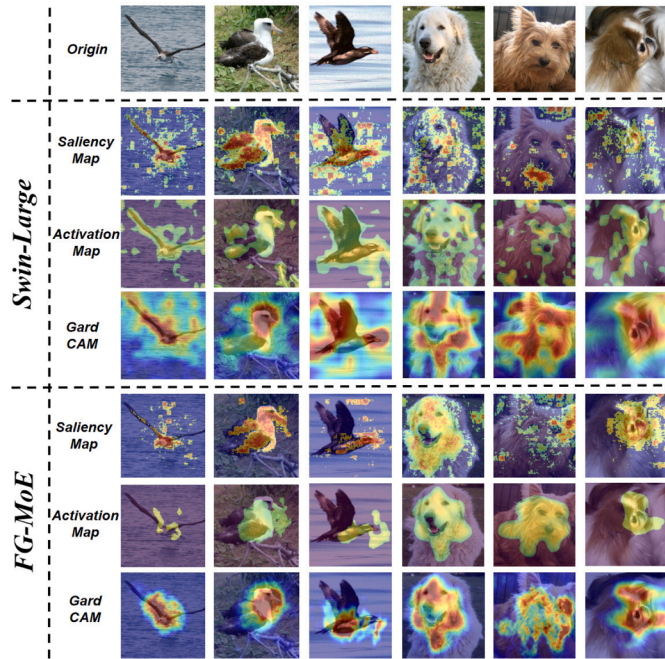


Fig. 7. Multiple visualization comparison between Baseline and FG-MoE.

Fig. 8 presents visualization of six test image groups selected from three datasets, with each image group subdivided into seven sub-images: one output image, attention maps from five experts, and one fused attention map determining the final classification.

Our MoE visualization decomposes gradient contributions from each expert module through hook mechanisms, revealing specialized attention patterns. As shown in Fig. 8, each expert demonstrates distinct focus areas: Global expert concentrates on overall object structure with some background context; Region expert emphasizes subject-background con-

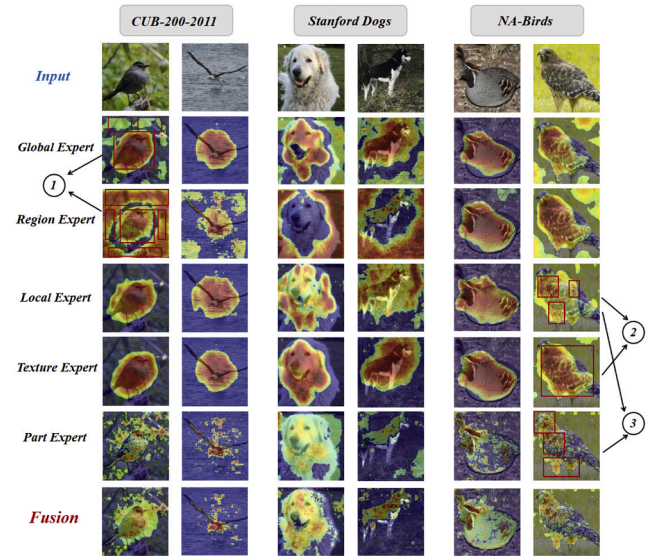


Fig. 8. Visualization of the focus areas of each expert and the fusion results.

trastive relationships; Local expert targets specific key areas; Texture expert captures multi-directional texture patterns; Part expert focuses on object components and inter-part dependencies.

Key distinctions include: ① Global vs. Region experts: Global focuses primarily on the main subject with minimal background, while Region extensively attends to subject-related background areas for contrastive analysis. ② Local vs. Texture experts: Local concentrates on small spatial regions, while Texture emphasizes rich textural information. ③ Local vs. Part experts: Part expert specifically targets anatomical components and their relationships, compared to Local's general area focus.

4.6.2. Visualization of expert allocation

Fig. 9 displays expert allocation for different test images. For the Cerulean-Warbler, the texture across its entire body represents a crucial feature; for the Northern-Flicker, a type of woodpecker, its dependency relationship with the background (such as trees in this instance) is equally an important reference worthy of attention; for this particular image of the Red-bellied-Woodpecker, its beak, head, body, and other parts are distinctly prominent, making it highly suitable for the part expert; while for test images like the Ivory-Gull where multiple characteristics are prominent, multiple experts participate more evenly in the decision-making process.

4.6.3. Confusion matrix visualization

We analyzed the 21 most frequently confused categories from CUB-200-2011 using confusion matrices for both Swin-L baseline and our FG-MoE model in Fig. 10. Bird names are abbreviated as initials for clarity (e.g., "Fish Crow" as "FC").

FG-MoE shows significant gains on challenging categories: Chuck-will's-widow accuracy rose from 0.66 to 0.90, Whip-poor-will from 0.58 to 0.93, Long-tailed-Jaeger doubled from 0.33 to 0.67. These improve-

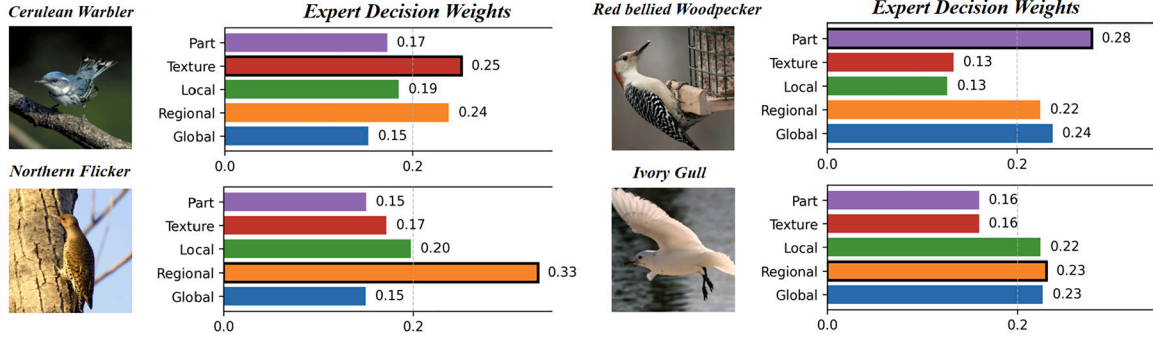


Fig. 9. Dynamic expert allocation for different images.

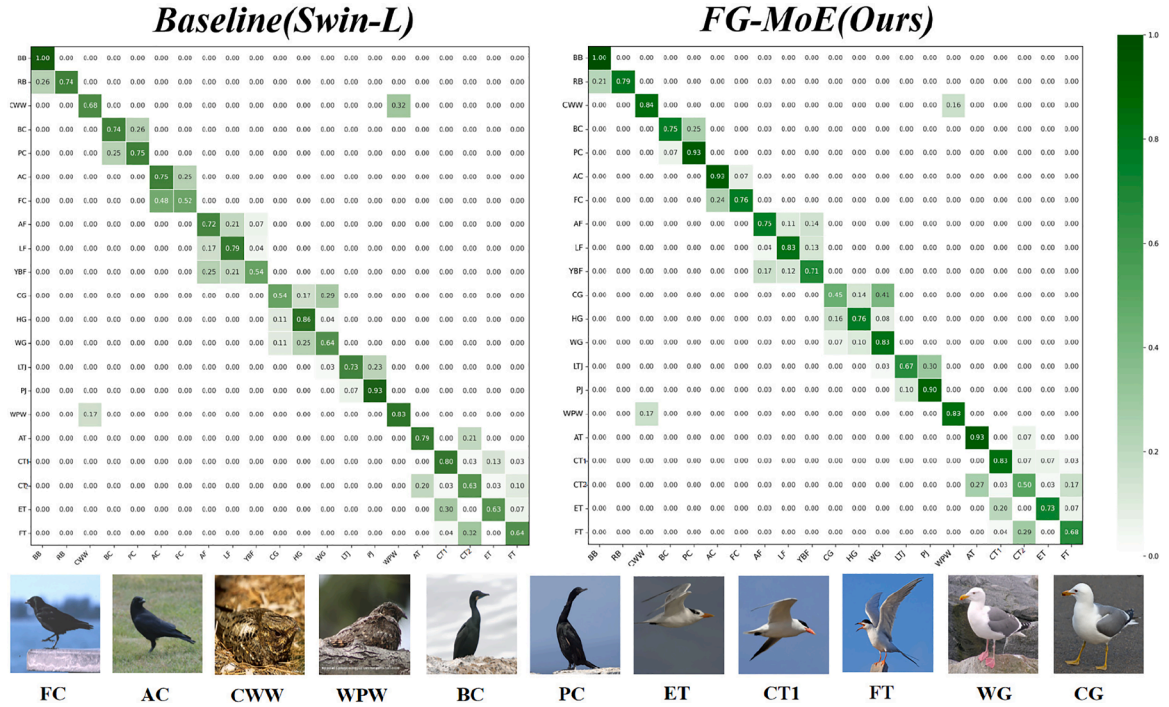


Fig. 10. Comparison of confusion matrices between Baseline and FG-MoE.

ments can be attributed to both the specialization of individual experts and their coordinated collaboration. Enhanced diagonal values and reduced off-diagonal confusion confirm our method's superior balance between accuracy and distinguishability. However, confusion between California-Gull(CG) and Western-Gull(WG) increased from 0.29 to 0.41. These species differ primarily in their leg coloration rather than shape or texture patterns. Our model excel at structural and textural discrimination but show limited sensitivity to subtle chromatic variations, suggesting future work should incorporate color-aware mechanisms for such cases.

For a comprehensive comparison, we evaluate the confusion matrices of multiple models-PMG [32], API-Net [31], ConvNeXt [18], ViT [6], Swin-B [11], TransFG [9], and Swin-L [11], using seven easily confused categories to enable clearer visualization in Fig. 11. Our FG-MoE demonstrates a marked improvement in distinguishing these challenging samples, significantly outperforming both CNN-based models and Transformer-based architectures. Fig. 11 also presents a t-SNE visualization comparing FG-MoE with three SOTA methods on most challenging samples. It can be observed that FG-MoE demonstrates superior discriminative capability for hard-to-distinguish cases.

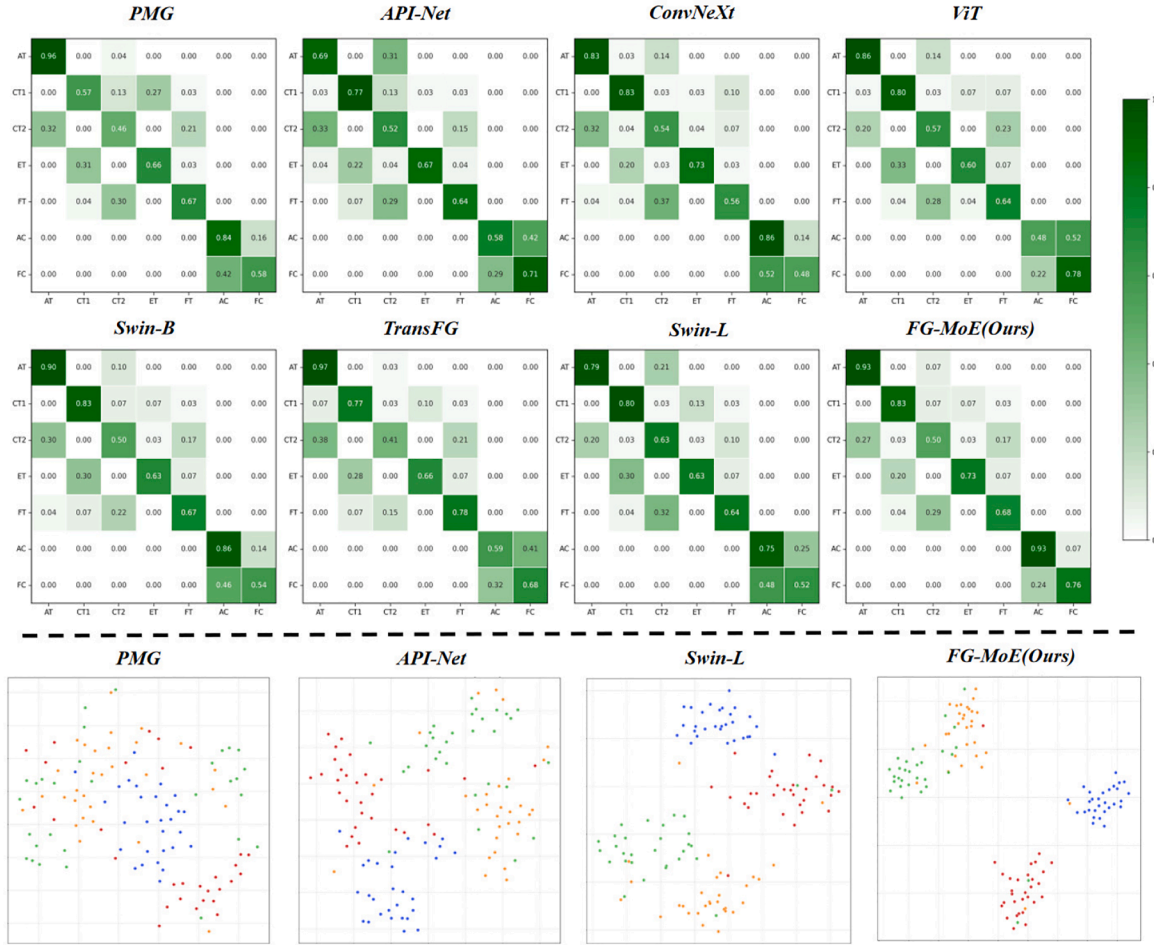


Fig. 11. Comparison of confusion matrices and t-SNE visualizations on the most challenging samples across different models.

5. Conclusion and limitation

In this paper, we propose a heterogeneous Mixture of Experts model, FG-MoE, to address the fine-grained visual classification problem. Building upon neuroscience foundations, this model extends and designs five heterogeneous experts-global, region, local, texture, and part-each focusing on distinct image features. We design a spatially-aware gating mechanism for expert allocation, complemented by diversity regulation, balance constraints, and multi-stage training strategies to jointly promote competition and collaboration among experts. Experimental results demonstrate that our model achieves transferability across multiple backbones with significant performance improvements, and attains state-of-the-art performance on multiple datasets.

While noticeable improvements are observable with FG-MoE on ResNet50, they have not surpassed all the state-of-the-art methods that also employ ResNet50 as their backbone. This is because the MoE/Attention enhancements are primarily optimized for the long-range dependencies in Transformer token sequences. The strong locality and aggressive downsampling of CNNs yield feature maps that lack the global correlations and fine details crucial for fine-grained visual classification, thus limiting the efficacy of our enhanced structures.

In future work, we aim to focus on designing MoE variants tailored for CNN features, and we plan to further investigate and optimize the dynamic competition-collaboration mechanisms among experts, introducing task-aware or class-aware soft routing strategies to enhance the model's robustness, especially for long-tail distributions and challenging samples.

CRedit authorship contribution statement

Songming Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization; **Jing Wen:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition; **Bin Fang:** Supervision, Project administration.

Data availability

The data that has been used is confidential.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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